LEPTOSPIRA PREVALENCE AND LINEAGES VARY ACROSS LAND USE TYPES DUE TO SHIFTS IN SMALL MAMMAL COMMUNITIES

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Introduction

Leptospirosis:

- Globally distributed, environmentally transmitted zoonosis, 2.9 million DALYs [1]
- Urine ⟷ Water/soil ⟷ Skin [2]

Madagascar:

- Species invasions [3] & deforestation [4]
- Distinct *Leptospira* lineages [5-6]

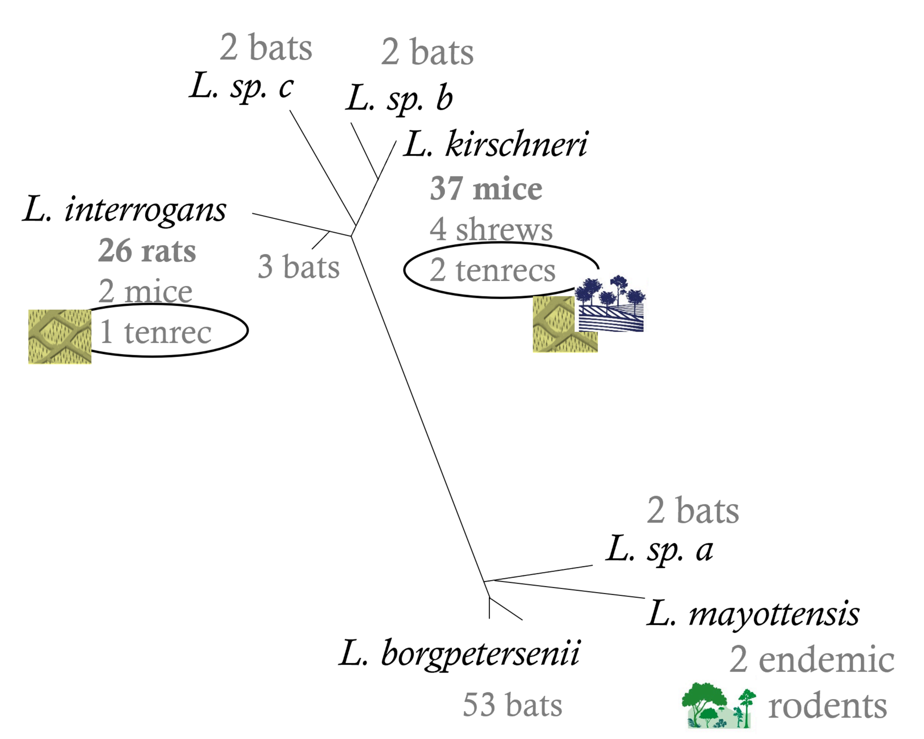

Figure 2.

Phylogenetic tree of *Leptospira* lineages by host and habitat types. Sequence groups into “Bat”, “Rat”, “Mouse”, and “Endemic small mammal” associated clades, except for the endemic small mammals captured in agricultural land use types.

Methods

1. Trap small mammals (Muridae, Soricidae, Tenrecidae, Nesomyidae) and bats in protected forests, agricultural land use types, & homes.
2. Test animals for *Leptospira* and determine *Leptospira* lineage using Sanger sequencing.
3. Assess host community composition (NMDS), *Leptospira* prevalence (GLMMs), and host-pathogen associations between land use types.

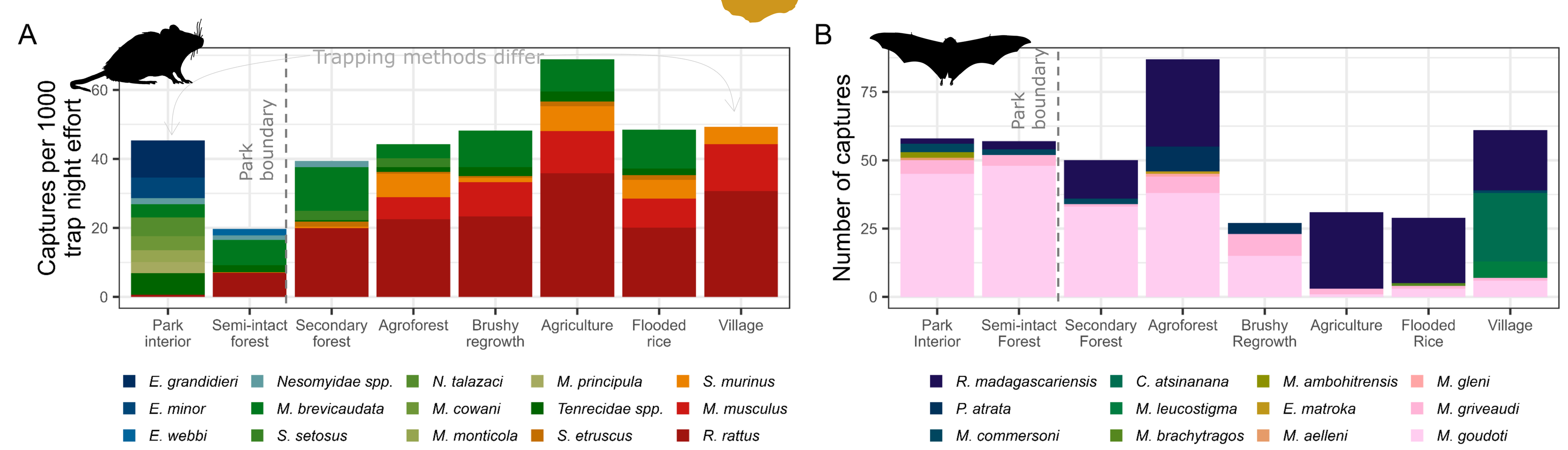

Figure 1.

Small mammal and bat species composition changes between land-use types, particularly inside the National Park boundary.

Objective

Investigate how land use correlates with the composition of host communities, *Leptospira* infection prevalence, and *Leptospira*-host species associations in terrestrial small mammals and bats across a land use gradient in rural northeastern Madagascar.

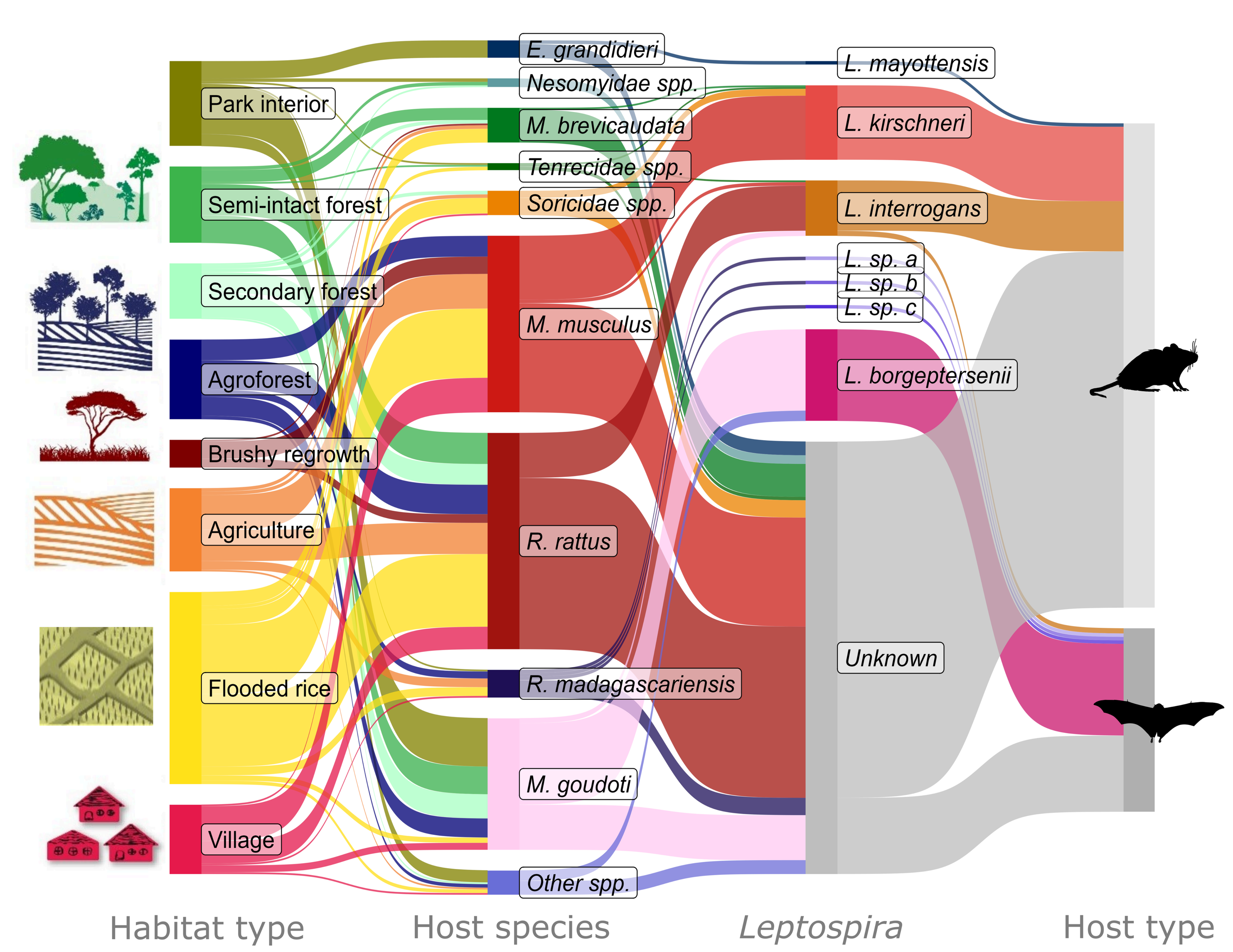

Figure 3.

Small mammal species composition & associated Leptospira infections vary with land use type.

Results

Small mammal communities differ between forested, agricultural, and village land-use types (n=2053; n spp = 28).

- PERMANOVA, density, richness, shannon

Bat communities differ in dominant species between land use types (n=400, n spp=12).

- PERMANOVA

13.6% of small mammals infected (280/2032).

- Mice>Rats>Shrews>endemic
- >50% of mice in rice fields
- Predictors: habitat, year, density (+)
- Sequenced 74 samples

37.7% of bats (106/281)

- *My. goudoti>R. madagascariensis*
- Forests > Ag.
- Predictors: Season/village
- Sequenced 62 samples

Sequencing results show host species cluster together. Three of the sequences from endemic small mammals captured in rice fields and secondary forests provide limited evidence of *Leptospira*-host shifts.

Conclusions

Agricultural land-use corresponds to the replacement of native species and endemic *Leptospira* lineages with non-native species and cosmopolitan *Leptospira* lineages. Together, contributing to higher infection prevalence in more disturbed habitats, particularly rice fields, providing evidence that land use results in changes in species composition that lead to novel host-parasite associations.

See our forthcoming publication in Applied Environmental Microbiology

Key Sources & Acknowledgements

[1] Costa et al., 2015. Global Morbidity and Mortality of Leptospirosis: A Systematic Review. PLOS NTD. [2] Levett. 2001. Leptospirosis. Clinical Microbio Reviews. [3] Goodman. 2022. The New Natural History of Madagascar. Princeton University Press. [4] Vieilledent, et al. 2018. Combining global tree cover loss data with historical national forest cover maps to look at six decades of deforestation and forest fragmentation in Madagascar. Biol. Conserv. [5] Rahelinirina, et al., 2019. High prevalence of Leptospira spp. in rodents in an urban setting in Madagascar. Am. J. Trop. Med. Hyg. [6] Gomard, et al., 2016. Malagasy bats shelter a considerable genetic diversity of pathogenic Leptospira suggesting notable host-specificity patterns. FEMS Microbiol. Ecol.

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